

TOI-2046b

R-2098

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_231027 · created 2026-08-10T23:01:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460244.59 +/- 0.15 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.046
Transit depth (Rp/Rs)^2	2.47 +/- 1.44 [%]
Orbital Inclination (inc)	87.88 +/- 2.89
Airmass coefficient 1 (a1)	0.763 +/- 0.011
Airmass coefficient 2 (a2)	0.188 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#1 - [279, 115]
Optimal Aperture	3.08
Optimal Annulus	3.62
Transit Duration (day)	0.034 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.046 vs 0.1213 (deviation 0.78σ)
Duration fit vs published	0.034 d vs 0.1033 d (deviation 1.69σ)
Mid-time O–C	-111.1 min
Depth SNR	3.4
Residual scatter	1.45 %

Light curve

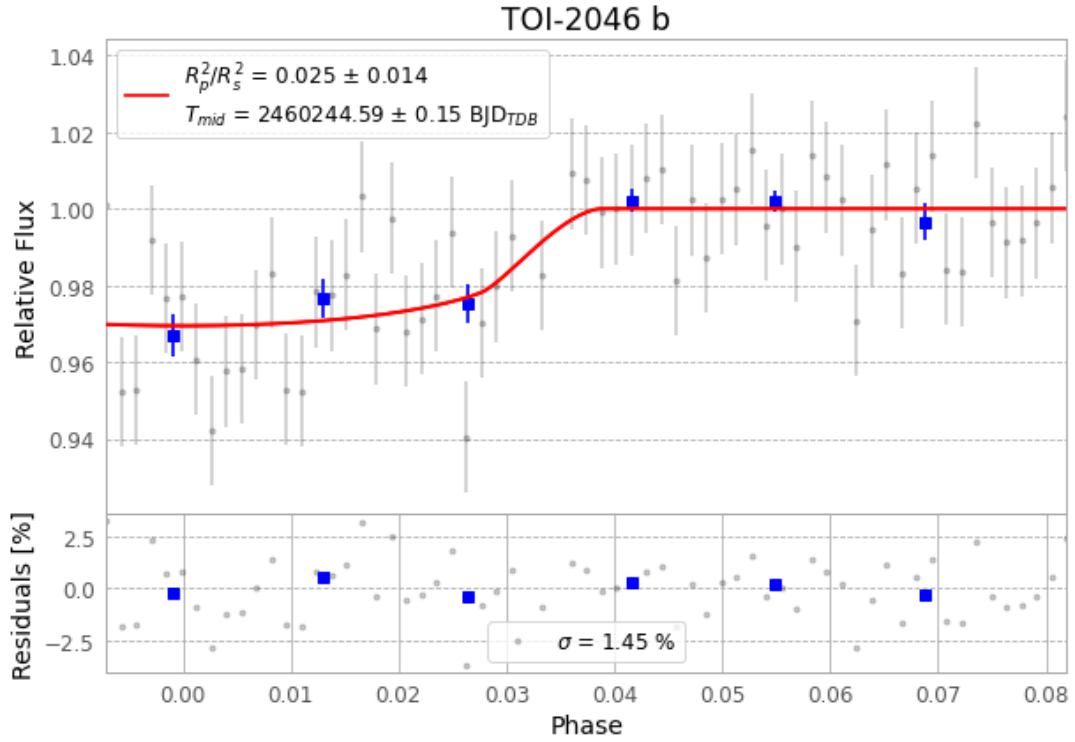


Figure 1 — FinalLightCurve_TOI-2046 b_2023-10-26.png (SHA-256 3a48a7d5b30d31e3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a30a881af6551570...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

65 FITS frames (43 MB), TOI2046231027014215.FITS → TOI2046231027045415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-231027.log (e3795a7c9e2d...), FinalLightCurve_TOI-2046 b_2023-10-26.png (3a48a7d5b30d...), FinalLightCurve_TOI-2046 b_2023-10-26.csv (1d4e8e6ef170...)

Compute resources

Wall time 121.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:01Z.